



Paper ID : 261852

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Subject Code:BP203T

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BPHARM
(SEM. II) THEORY EXAMINATION 2025-26
BIOCHEMISTRY

TIME: 3 HRS**M.MARKS: 75**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt all questions in brief.

10 x 2 = 20

a.	Write a note on the hormonal regulation of blood glucose level.
b.	Define genetic code with example.
c.	Enlist the enzymes involved in DNA replication and state their functions.
d.	Define isoenzymes and give example.
e.	Define atherosclerosis and fatty liver.
f.	Define free energy change (ΔG).
g.	What is G6PD deficiency?
h.	Explain ketone bodies with their structures.
i.	Explain endergonic reactions and exergonic reactions.
j.	Define glycogenesis and glycogenolysis.

SECTION B

2. Attempt any two parts of the following:

2 x 10 = 20

a.	What is HMP shunt? Explain its reactions. Write biological importance of this pathway.
b.	Critically evaluate the role of biomolecules in maintaining cellular structure and metabolic functions.
c.	Derive the Michaelis–Menten equation and explain the significance of V_{max} and K_m in enzyme kinetics.

SECTION C

3. Attempt any five parts of the following:

7 x 5 = 35

a.	Discuss the biological significance of cholesterol and its conversion into bile acids and vitamin D.
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b.	Explain the properties of enzymes and factors affecting enzyme activity.
c.	Discuss the formation, utilization and physiological significance of ketone bodies. Explain ketoacidosis with suitable examples.
d.	Briefly explain De novo synthesis of fatty acids.
e.	Discuss the catabolism of purine nucleotides and explain the biochemical basis of hyperuricemia and gout disease.
f.	Describe Electron transport chain (ETC) and its mechanism.
g.	Develop a comparative analysis of carbohydrates, lipids and proteins with respect to structure, energy yield and physiological functions.